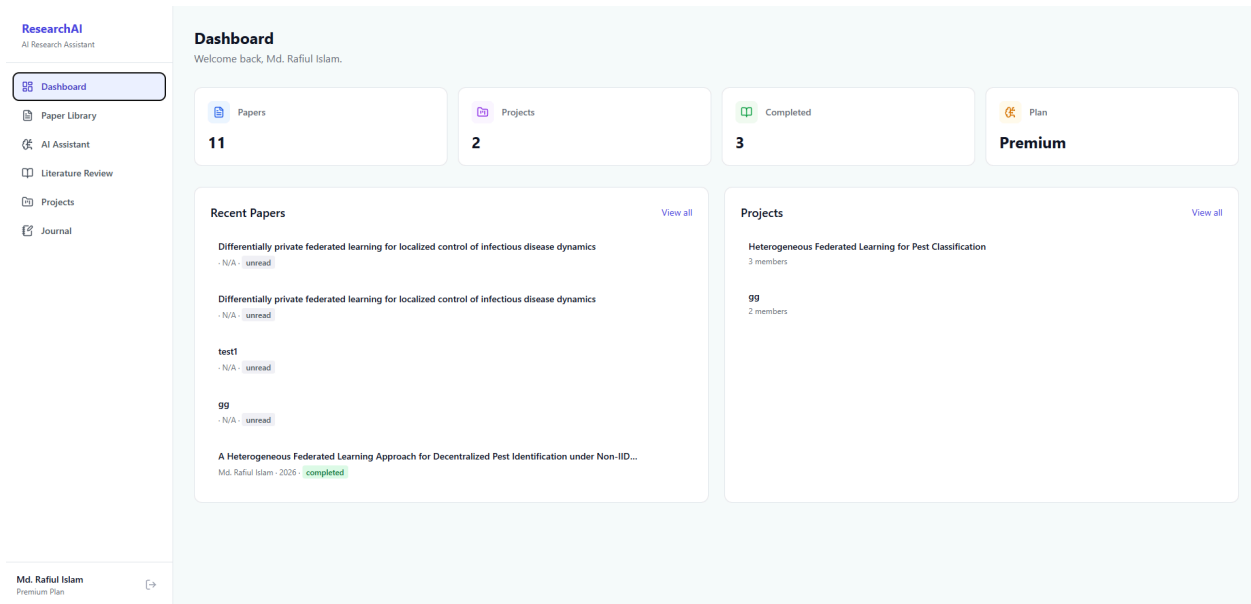
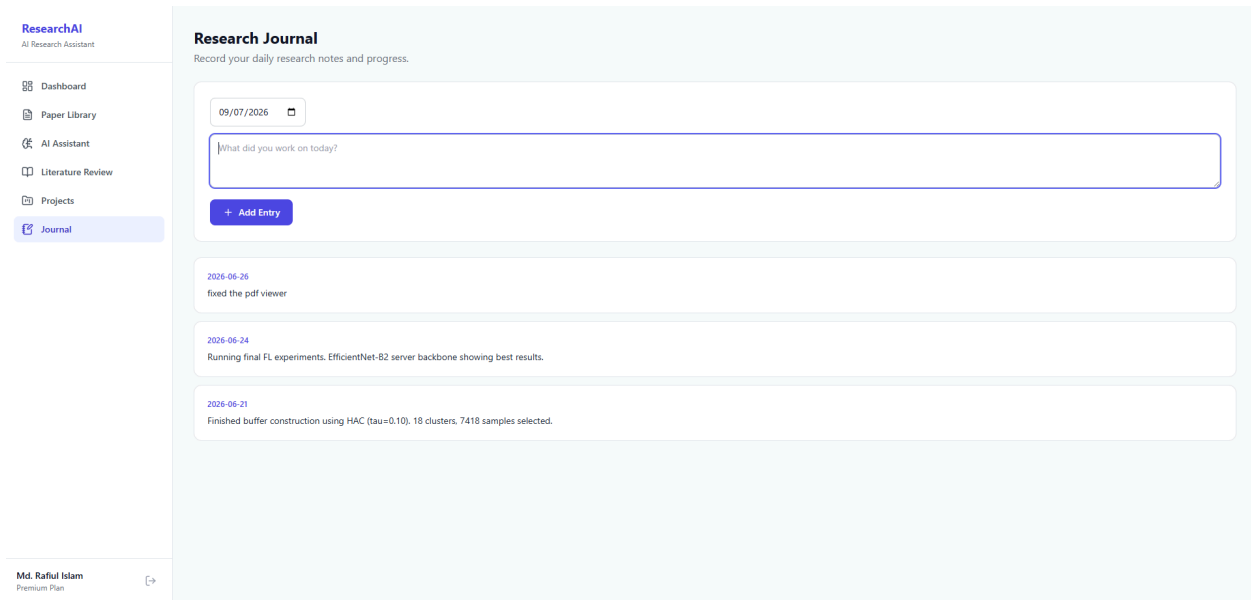


Rafit

Feature-1: dashboard

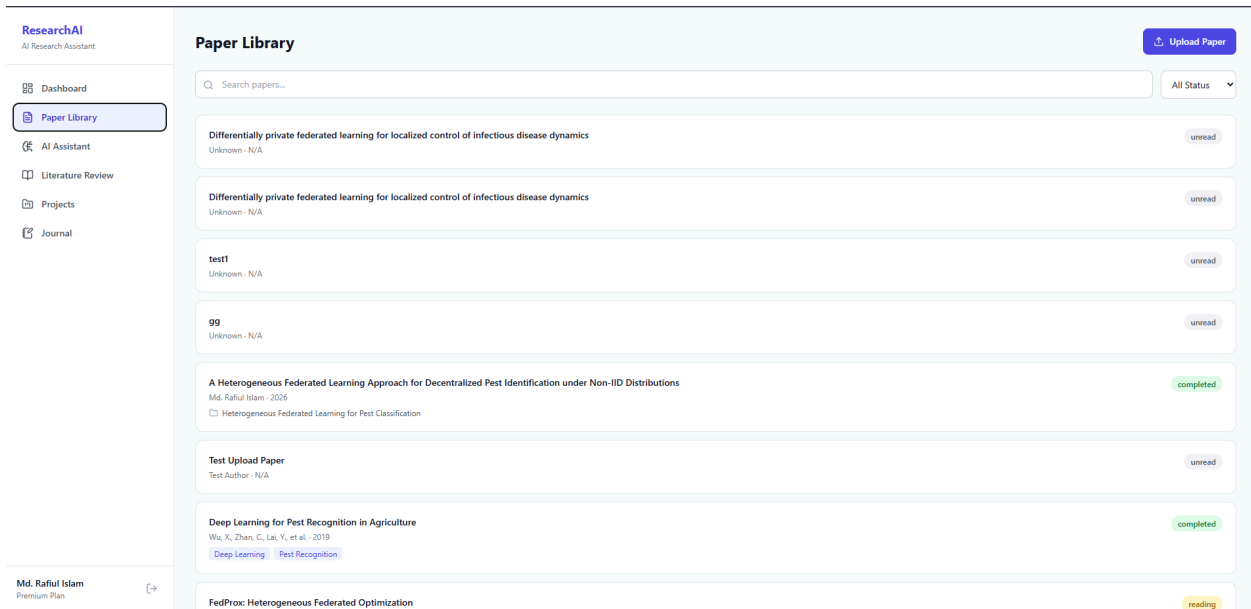


Feature-2: Journal

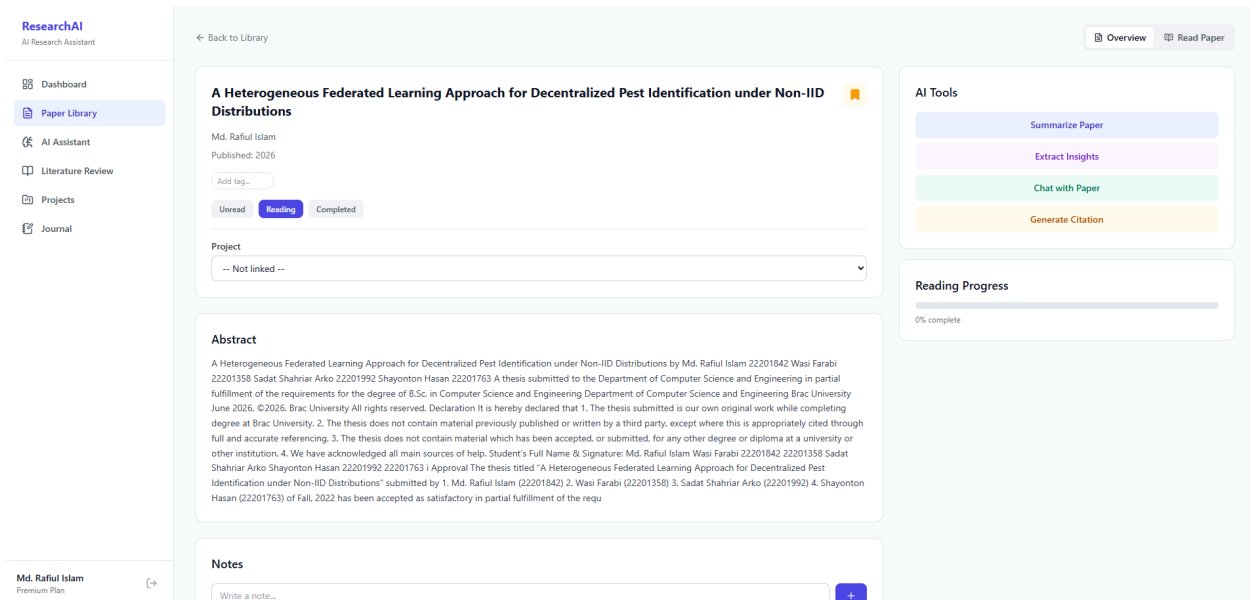


Reaz

Feature-1: paper library



Feature-2: Paper overview (ignore the AI tools on the right for now)



Wasi

Feature-1: AI assistance (summarize paper)

ResearchAI

AI Research Assistant

Dashboard

Paper Library

AI Assistant

Literature Review

Projects

Journal

Md. Raful Islam

Premium Plan

AI Research Assistant

Use AI to analyze papers, find gaps, and generate ideas.

Summarize Paper

Extract Insights

Chat with Paper

Research Gap Finder

Research Questions

Title Generator

Summarize Paper

Select Paper

A Heterogeneous Federated Learning Approach for Decentralized Pest Identification under Non-IID Distributions

Run

Result

Copy JSON

Abstract Summary

This thesis introduces a heterogeneous Federated Learning (FL) framework designed for fine-grained agricultural pest classification, specifically addressing challenges posed by non-IID data distributions and gradient-based privacy attacks. The approach utilizes diverse client architectures and aggregates only a shared classifier head to prevent direct gradient comparisons. It integrates a server-maintained clustered spatial buffer and server-anchored knowledge distillation to improve learning under data heterogeneity. This framework aims to enhance classification performance and privacy protection without relying on traditional privacy mechanisms.

Key Findings

The experimental results demonstrate that the proposed heterogeneous FL framework achieves a test accuracy of 63.76% and an F1-score of 0.611. This performance significantly outperforms the heterogeneous baseline by 4.16 percentage points. Crucially, the framework exhibits strong privacy protection by reducing property inference attack accuracy from 56.7%-61.7% in a homogeneous FL baseline to 0%. These findings underscore the effectiveness of server-anchored knowledge distillation in mitigating client divergence, enhancing classification performance under non-IID conditions, and structurally improving privacy without requiring differential privacy or secure multi-party computation.

Methodology Summary

The proposed methodology employs a heterogeneous FL framework with diverse client model architectures, applied to the IP102 dataset partitioned using a Dirichlet distribution ($\alpha=0.5$) to simulate non-IID conditions. Key to its design is partial model aggregation, where only the shared classifier head is aggregated, preventing full gradient exposure across clients. To address data heterogeneity, a server-maintained clustered spatial buffer is constructed using Hierarchical Agglomerative Clustering. This buffer generates class-prior knowledge, which is then utilized for server-anchored knowledge distillation, progressively strengthened throughout the training process.

Feature-2: AI Assistance (Title Generator)

ResearchAI

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Research Questions

Title Generator

Title Generator

Research Description

Human-Computer Interaction (HCI) is an interdisciplinary field studying how people interact with technology. It draws on computer science, psychology, and design to create intuitive, accessible systems. A core area of research is Usable Privacy and Security, which designs technologies that help users protect their data without needing advanced technical expertise

Run

Result

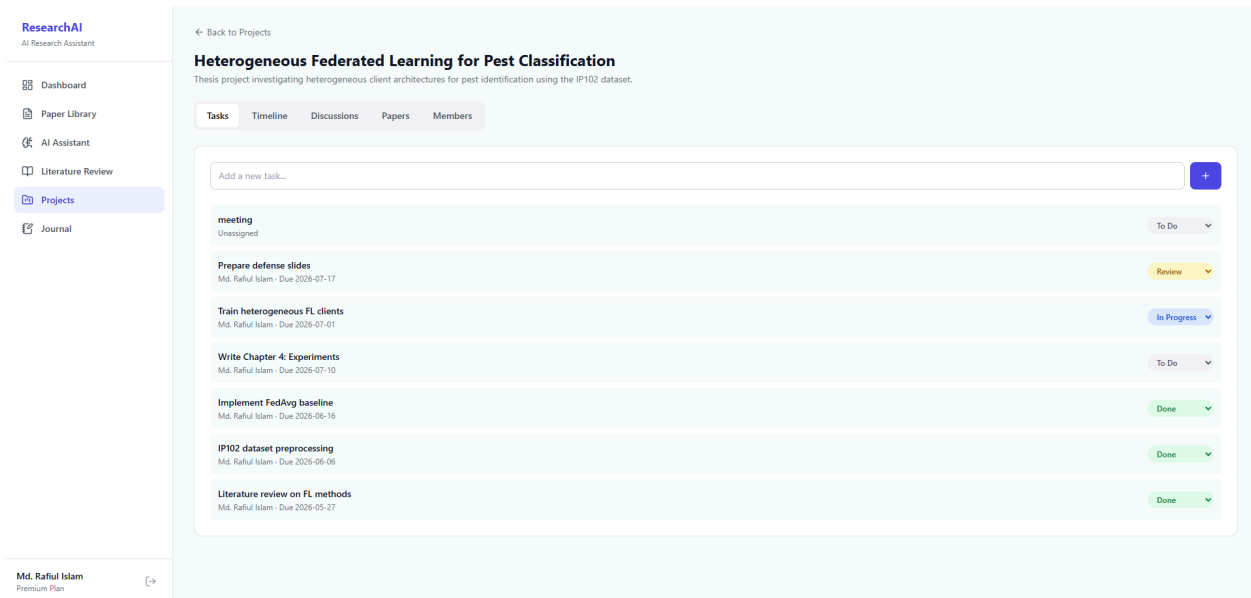
Copy JSON

Titles

- Usable Privacy and Security in Human-Computer Interaction: A Framework for Intuitive Data Protection
- The Human Shield: Building Instinctive Privacy and Security in the Digital Realm
- Designing for the Uninitiated: Enhancing Data Protection Through Usable HCI Principles
- Unlocking Peace of Mind: HCI Strategies for Effortless User Privacy and Security
- An Interdisciplinary Approach to Usable Security and Privacy: Integrating Psychology, Design, and Computer Science

Naina

Feature-1: Project task progress tracking



Feature-2: Discussion with other project members and Supervisor

